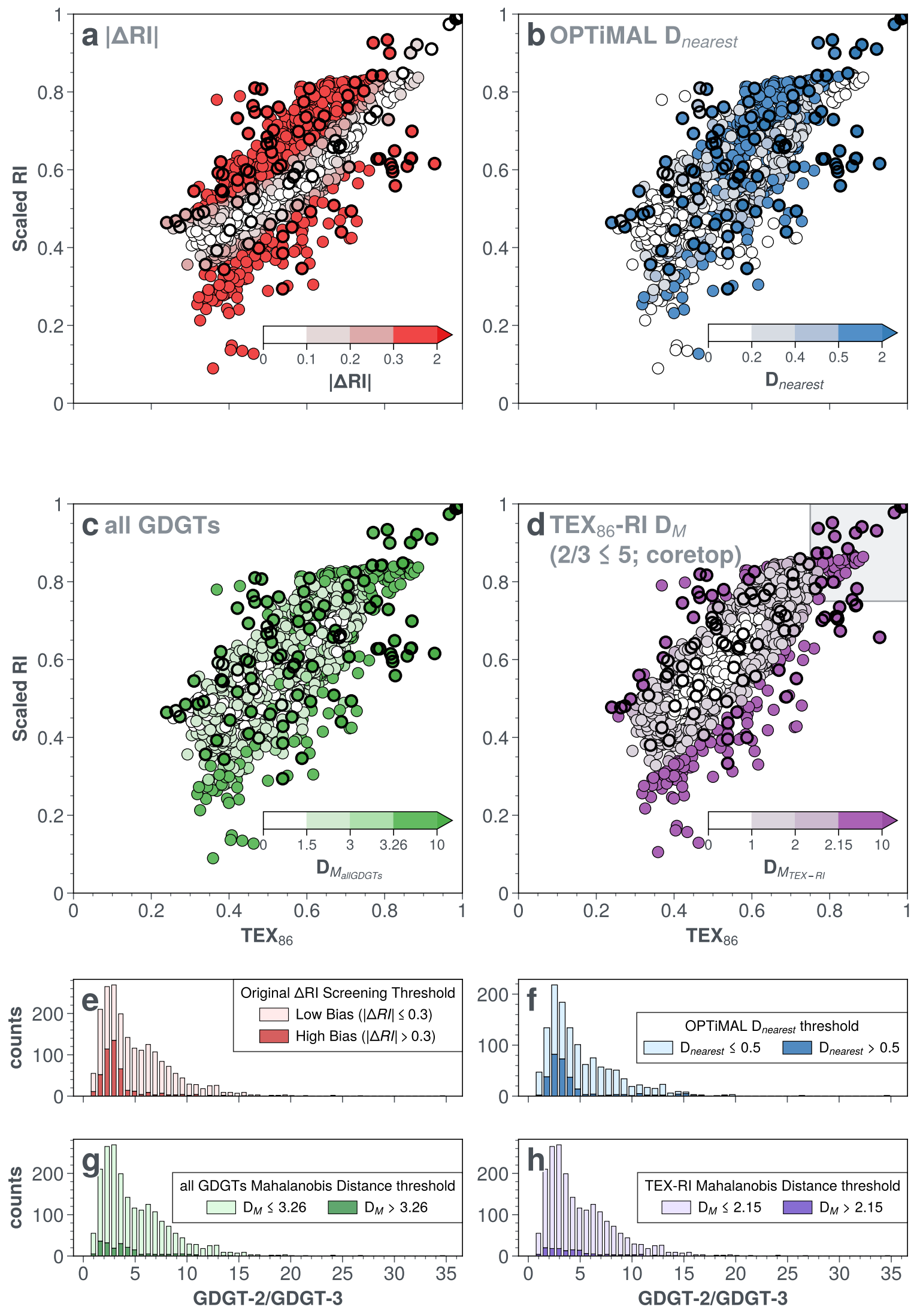

a |ΔRI|
Scaled RI
0 0.1 0.2 0.3 2
|ΔRI|
b OPTiMAL D nearest
0 0.2 0.4 0.5 2
D nearest
c all GDGTs
0 1.5 3 3.26 10
D M allGDGTs
TEX86
d TEX86-RI D M
(2/3 ≤ 5; coretop)
0 1 2 2.15 10
D M TEX−RI
TEX86
counts
e
Original ΔRI Screening Threshold
Low Bias (|ΔRI| ≤ 0.3)
High Bias (|ΔRI| > 0.3)
f
OPTiMAL D nearest threshold
D nearest ≤ 0.5
D nearest > 0.5
g
all GDGTs Mahalanobis Distance threshold
D M ≤ 3.26
D M > 3.26
GDGT-2/GDGT-3
h
TEX-RI Mahalanobis Distance threshold
D M ≤ 2.15
D M > 2.15
GDGT-2/GDGT-3